Ryan Sharr

AI SYSTEMS ENGINEER / APPLIED AI ENGINEER

Remote (US)
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SUMMARY

AI systems engineer building pipelines that convert prompt-driven input into controllable, stateful runtime systems. Focused on simulation architecture, state ownership, and making AI output inspectable, debuggable, and verifiable as real software.

Background in game-engine systems, with transferable architecture across simulations, tooling, and applied AI products.

CORE SKILLS

AI Systems: LLM pipelines (structured systems → runtime), structured prompting, rapid prototyping

Architecture: simulation systems, state ownership, command-driven runtime control

Runtime Systems: stats, conditions, behaviors, state machines

Pipelines: import → validate → build, data-driven systems

Engineering: Unity / C# architecture, applied AI, product systems

Observability: logging, validation, inspectable runtime state

SECONDARY SKILLS

3D modeling and environment art

3D animation and asset integration

Concept art interpretation and production

UI/UX and gameplay readability

Game design docs and data structures

Content pipelines and asset workflows

TECHNICAL STACK

Languages: C#, JavaScript

Engines / Tools: Codex, Unity, Unreal Engine, Godot, UEFN (Verse), Custom Tooling

Systems: Runtime simulation, gameplay systems (stats, conditions, buffs), state machines, character controllers, navigation & pathfinding, AI behavior systems, data-driven architectures, databases, tooling & editor systems

Art / Web: 3ds Max, Maya, ZBrush, Photoshop, Git, HTML/CSS

EDUCATION

BFA – Media Arts & Animation
Art Institute of Seattle

SELECTED WORK

AI → Playable Runtime System

ryansharr.com

PROMPT TO RUNTIME PIPELINE

End-to-end pipeline converting prompt-driven input into structured systems, generated content, and controllable runtime behavior.

- Prompt → structured data → generated systems → runtime simulation
- Inspectable runtime state with debugging and validation tooling
- Built around structured runtime authority, not one-off generation
- Demonstrates AI output as controllable, stateful software
- Designed around verifiable system behavior, not static generation

Complex Shooter Systems

- Runtime stats, buffs, conditions, and gameplay behavior.
- Live editing for resources, conditions, and internal state.
- Cross-system sync across AI, items, economy, and definitions.
- State-driven architecture for coordinated runtime systems.

3D Pathfinding Prototype

- AI-managed room generation with volumetric node navigation.
- Spatial traversal across walls and ceilings.
- Geometry-derived traversal graph baking.
- AI-driven environment generation and navigation.

EXPERIENCE

Engineering Lead / Creative Director

Loot Foundry Ltd

Led engineering and creative direction for system-driven gameplay development in UEFN (Verse), bridging design, art, and implementation to deliver structured, reusable runtime systems.

- Built core gameplay systems in Verse including stats, conditions, buffs/debuffs, and state-driven behavior.
- Designed runtime architecture with centralized logic decoupled from scene objects.
- Directed system design across gameplay, UX, and technical implementation.
- Partnered across disciplines to translate concepts into structured systems.
- Served as primary technical point of contact, debugging and unblocking workflows.
- Developed reusable systems for gameplay state and cross-system interaction.

Independent Systems Development

Self-directed

Transitioned from 3D character art into full-stack systems, simulation architecture, and game development.

- Self-taught programming and system architecture from the ground up.
- Built end-to-end gameplay systems including state management and runtime behavior.
- Designed complete systems spanning engineering, design, content, and tooling.
- Built pipelines that connect structured input to runtime behavior and observable output.
- Developed deep understanding of system interaction, tooling, and runtime pipelines.

3D Character Artist

Gas Powered Games

Worked on Age of Empires Online.

- Modeled and textured production-ready character assets from concept art.
- Collaborated with animation, VFX, and design for gameplay integration.
- Worked within production pipelines under technical and performance constraints.
- Produced UI-supporting visual assets.